

# Earned Business Value

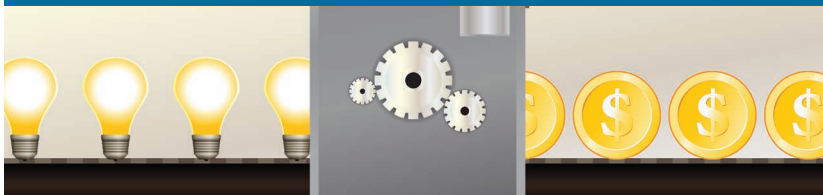
## See That You Deliver Value to Your Customer

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// A bundle of easy-to-use core practices help developers systematize a project's evolving knowledge. Although these practices are simple, they provide powerful support for monitoring project progress on both the realized business value and the incurred costs. //



**THE ORDER IN** which you send your backlog items into construction determines when stakeholders can reap benefits from a particular piece of functionality. This can substantially impact market timing, enterprise earnings, and the project manager survival rate. There are several ways to order a backlog, and sophisticated methods and tools exist to do so—for example, during release planning.

But no matter what backlog-ordering scheme you use, you ought to be explicit on the order in which you realize potential business value.

To that end, we present methods to express business value relative to cost in your backlog. We also provide methods to monitor how much potential business value you're realizing along the way—in addition to the cost expended.

### Assigning Benefit Points

Given business value's central role in agile development, we previously said that you should assign *benefit points* (BPs) to your project's product elements (epics and stories), with at least the same vigor and rigor with which you assign story points (SPs) for cost.<sup>1</sup> (Epics are high-level requirements perceived at the project's early stages, typically at project inception. In contrast, stories are more refined and elaborated user stories reflecting the project's understanding at more advanced stages.) To do this, assign BPs (for example, in a benefit poker session using the Fibonacci sequence) according to how much you think an epic contributes to the project's distinct objectives (see Figure 1).

Objectives, which are part of the project's business case, express the desired effect of the project's deliverables on the organization. The objectives might, in turn, be assessed to determine to what degree they contribute to the enterprise's planned returns. Then, to reflect that objectives might contribute unevenly to return, the BPs are balanced accordingly. All this constitutes the core practice of assigning BPs for epics.<sup>1</sup>

Consider an example from the public-service domain that involves three objectives:

- Objective 1. Reduce the average case-processing time by 30 percent.
- Objective 2. Reduce the number of wrong case decisions by 90 percent.
- Objective 3. Reduce the average interaction time between the applicant and application processor by 70 percent.

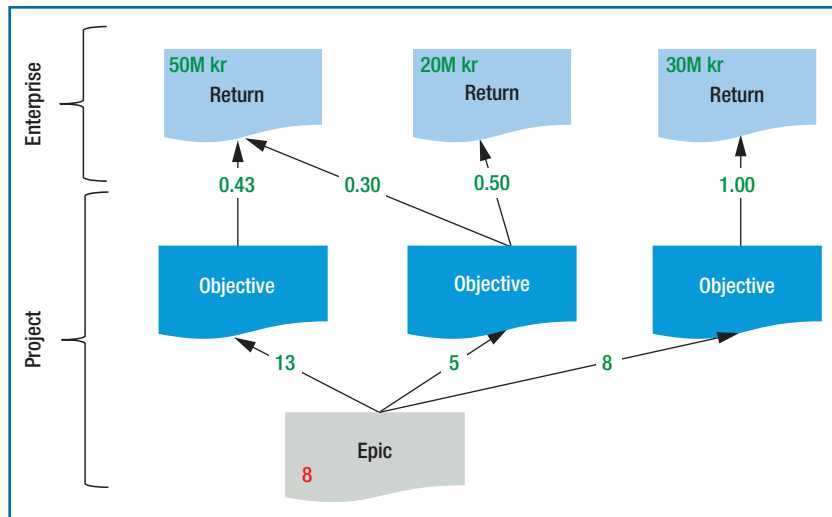
We have three returns:

- Return 1. Reduced number of person-hours—worth 50 million Norwegian kroner (US\$1 = 8.5 kr).
- Return 2. Reduced monetary compensation—worth 20 million kr.
- Return 3. The organization's improved public image—worth 30 million kr.

There are eight epics; here are four of them:

- Epic 1. **As the applicant I can** secure my identity in the application process **by using** the system to authenticate myself ....
- Epic 2. **As the applicant I can** start with a prefilled application form **by using** the system to retrieve and autofill all available and relevant information ....
- Epic 3. **As the case processor I can** find all the relevant information for a case **by using** the system to retrieve applicant information from all relevant and permissible data sources in a single search ....
- Epic 4. **As the division manager I can** manage my division's productivity **by using** the system to view statistics to monitor the time and quality of case processing ....

Figure 2a shows the first estimation task for this project: estimate the business value for epics in the form of BPs. For example, using the Fibonacci sequence, we estimate that Epic 1 contributes 13 BPs to Objective 1, 5 BPs to Objective 2, and 8 BPs to Objective 3, for a total of 26 BPs. The total BPs for all the epics is 211. (In Figure 2, your project stakeholders must produce the estimates in the white background, while those in the shaded background



**FIGURE 1.** An epic with its associated story point (SP—in this case, the red 8) and benefit points (BPs—in this case, the green 13, 5, and 8). The numbers above the objectives indicate to what degree those objectives contribute to the returns. Each return has a monetary value; for example, the first return has a value of 50 million Norwegian kroner (US\$1 = 8.5 kr).

should be automatically generated by your tool.)

Figure 2b shows the second estimation task: estimate how much each objective contributes to returns. The total strategically planned return for Returns 1, 2, and 3 is 100 million kr. We estimate that Objectives 1, 2, and 3 will contribute \$21.5 million, \$25 million, and \$30 million, respectively, to those returns—in all, \$76.5 million. So, the project's objectives, once fulfilled, will contribute unevenly to the project's return and only partly to the enterprise's strategically planned return.

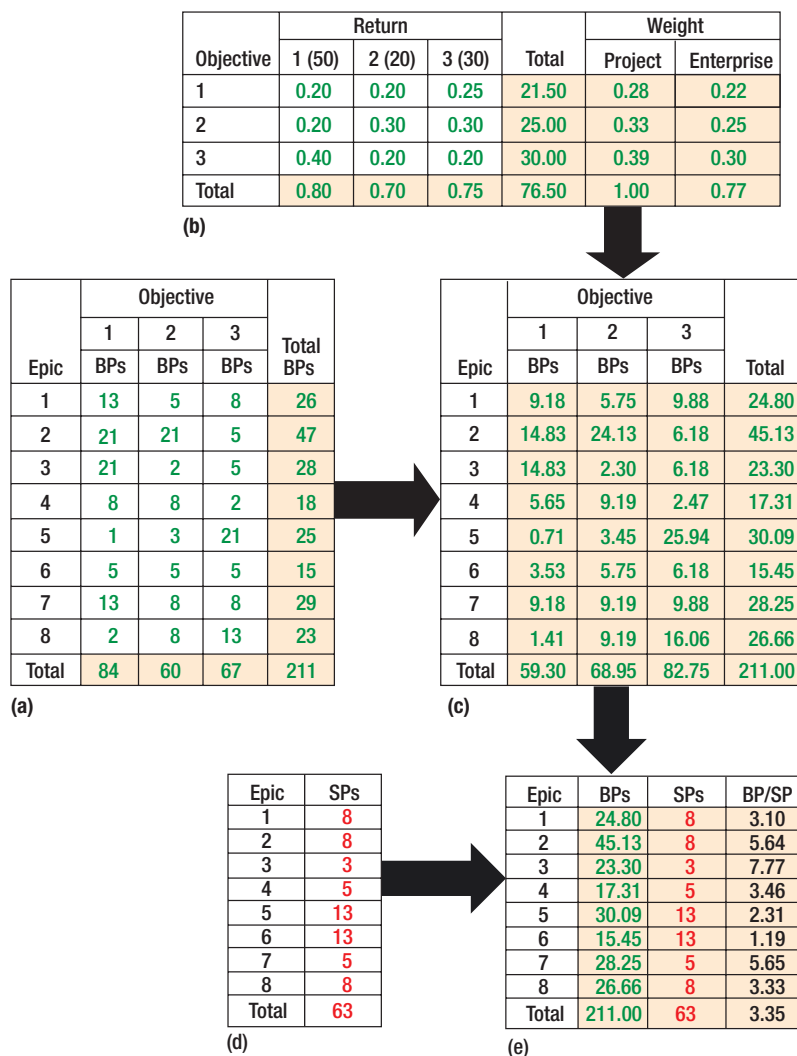
Figure 2c shows the BPs automatically balanced to account for objectives with different values.<sup>1</sup>

This core practice of assigning BPs effectively links the project's product estimates to the business case and to strategic plans. The aim is to harness and systematize stakeholders' insights and project learning, rather than to employ sophisticated tools to

calculate estimates that fare no better on average than expert estimation.<sup>2</sup> So, methods must be simple, support experts' cognitive processes, and give sufficient, rather than optimal, results.<sup>3</sup> Although this core practice is simple, it supports powerful monitoring techniques.

You can use BPs along with SPs to obtain means to monitor and adjust your project. Assigning SPs (another core practice) routinely occurs in projects, so we'll assume you know how to do it—for example, in planning-poker sessions. However, we'll make a few remarks in the context of combining benefit and cost management.

Benefit manifests itself after deployment. So, to get a sensible combined measure of benefit and cost, cost estimates should include the postdeployment cost in addition to the development cost. Traditionally, SPs reflect only the development cost. However, it's common to assume that the lifecycle cost is



**FIGURE 2.** Assigning BPs and SPs for epics. (a) The epics' contribution to objectives. (b) The objectives' contribution to returns. The numbers in parentheses indicate the expected overall returns in millions of kroner. For each objective, we obtained the sum in the "total" column to the right of the "return" column by multiplying those goals by that objective's expected contributions and summing the results. For example, for objective 1,  $(50 \times 0.20) + (50 \times 0.20) + (50 \times 0.25) = 21.50$ . (c) The resulting balanced BPs. (d) The SPs for lifecycle cost. (e) The BP/SP ratio. Green indicates benefit; red indicates cost. Your project stakeholders must produce the estimates in the white background, while those in the shaded background should be automatically generated by your tool. The numbers are presented in two-decimal precision.

assumption, you can assign SPs as you did before; an SP's monetary value is what determines whether it reflects the lifecycle cost or just the development cost. Our methods apply regardless of that assumption, but if you do make that assumption (and it's warranted), some of the methods can take on a simpler form. For our running example, we assume the SPs in Figure 2d.

We then calculate the BP/SP ratio in Figure 2e to get a relative benefit/cost measure. You can divide BPs by SPs because both types of points are on a ratio scale. If you use categorical or nominal schemes—say, MoSCoW,<sup>5</sup> which gives you four categories of importance (must have, should have, could have, and won't have)—to assess benefit, you can't divide your benefit estimates by cost. To get a benefit/cost measure out of MoSCoW, you could rank the product elements according to their decreasing cost in each category and then order the backlog by selecting the ranked elements successively, starting with must have, then should have, then could have, then won't have. However, one element in a less important category might have a higher actual benefit/cost ratio than another element in a more important category, owing to low cost. Without a sound measure of benefit/cost provided by ratio scales, you wouldn't become aware of such incidents.

## Applying the Core Practice

We now show how to apply BP estimation to three project management activities.

### Ordering the Epic Backlog

Given BPs and SPs for epics, you can start making informed decisions on your backlog's order. Several tactics

proportional to, or linearly dependent on, the development cost—for example, depending on the domain and type of system.<sup>4</sup> Under that

are possible; we'll look at one in detail and discuss others as we go along.

Table 1 is Figure 2e sorted according to the decreasing BP/SP ratio, which is the main order in which you should develop and elaborate epics. This maximizes the business value for the customer, relative to cost, early. (Other factors are also relevant to the order—for example, dependencies, risk, and uncertainty. We comment on dependencies later.)

This differs markedly from maximizing business value at all costs and differs from maximizing business value within a fixed cost or schedule bound. The idea here is that exceeding the budgeted cost might be okay if your estimates show you'll generate benefit that justifies that cost; that is, if the benefit/cost ratio is sufficiently high. Just as relevant, it's okay to stop construction when the benefit/cost ratio falls below a certain level. The project's ultimate goal is to fulfill its objectives, thereby realizing business value, as long as this doesn't cost more than it gains.

The iron triangle of project management (see Figure 3a) considers quality to be the result of balancing scope, schedule, and cost. The agile community has debated this concept and perhaps abandoned it for the agile triangle (see Figure 3b). In the agile triangle, the business value, or benefit, is a prime factor, in recognition of the prime objective to deliver valuable software to the customer. Perhaps because trying to balance factors inadvertently leads to one of them (for example, cost) becoming a goal in itself, rather than a means to achieve quality, the agile triangle has quality as a distinct factor.

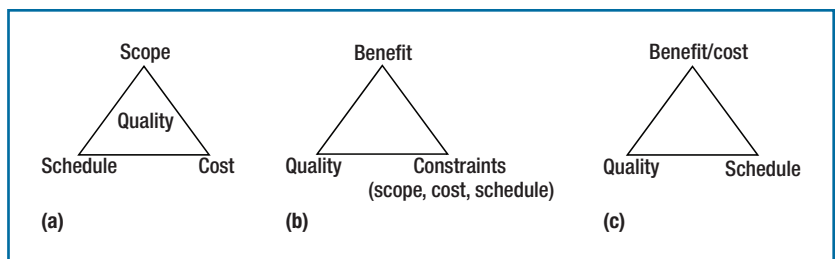
Although the iron and agile triangles aim to strike a balance between their factors, the intrinsic polarization makes it tempting to

TABLE 1

The benefit/cost ratio in terms of points.\*

| Epic  | BPs    | SPs | BP/SP |
|-------|--------|-----|-------|
| 3     | 23.30  | 3   | 7.77  |
| 7     | 28.25  | 5   | 5.65  |
| 2     | 45.13  | 8   | 5.64  |
| 4     | 17.31  | 5   | 3.46  |
| 8     | 26.66  | 8   | 3.33  |
| 1     | 24.80  | 8   | 3.10  |
| 5     | 30.09  | 13  | 2.31  |
| 6     | 15.45  | 13  | 1.19  |
| Total | 211.00 | 63  | 3.35  |

\* Green indicates benefit; red indicates cost. BP = benefit point; SP = story point.



**FIGURE 3.** The three triangles of project management. (a) The iron triangle. (b) The agile triangle. (c) The benefit/cost triangle. The benefit/cost triangle takes into account that benefit and cost shouldn't be polarized but should be integrated into a single metric.

emphasize one factor over the others. We contend that benefit and cost shouldn't be polarized but should be integrated into a single metric. So, the benefit/cost triangle (see Figure 3c) has the benefit/cost ratio as a factor. Quality refers to technical quality, including architecture, and the schedule is the remaining constraint because the scope is really a part of benefit. Here, we look at maximizing the benefit/cost ratio (subject to the schedule and quality).

A powerful aspect of using relative sizes such as BPs and SPs is that

you can assign monetary values to your points, according to current knowledge. The initial values you set, before project learning, will be based on the project's business case.

For example, Table 2 shows the same information as in Table 1, but with a BP's monetary value (0.36 million kr) set as the project's total estimated return (76.50 million kr) divided by its total BPs (211). In addition, an SP's lifecycle cost is 0.60 million kr, based on, say, structured stakeholder meetings and past experience from earlier projects with

TABLE 2

The benefit/cost ratio in millions of kroner.\*

| Epic  | Benefit | Cost  | Benefit/cost |
|-------|---------|-------|--------------|
| 3     | 8.45    | 1.80  | 4.69         |
| 7     | 10.24   | 3.00  | 3.41         |
| 2     | 16.36   | 4.80  | 3.41         |
| 4     | 6.28    | 3.00  | 2.09         |
| 8     | 9.67    | 4.80  | 2.01         |
| 1     | 8.99    | 4.80  | 1.87         |
| 5     | 10.91   | 7.80  | 1.40         |
| 6     | 5.60    | 7.80  | 0.72         |
| Total | 76.50   | 37.80 | 2.02         |

\* 1 BP = 0.36 million, 1 SP = 0.60 million.

similar characteristics. For example, structured discussions might have established an SP's development cost as 0.30 million kr. Then, a linear model of postdeployment cost might suggest that the lifecycle cost is twice that of the development cost. So, the lifecycle cost estimate is 37.80 million kr, and the lifecycle benefit estimate is 76.50 million kr.

With monetary values, benefit and cost have the same denomination. As Table 2 shows, Epic 6 has a benefit/cost ratio below 1, which means that it shouldn't be put into construction as a whole because it will return less benefit than it costs.

### Ordering the Story Backlog

Suppose that the project deploys in releases and that Epics 3, 7, and 2, in harmony with our tactic, are wisely planned for the first release. In line with just-in-time detailing, this is the point at which to elaborate those epics into stories.

So suppose Epics 3, 7, and 2 are elaborated into stories (see Table 3).

Now, BPs and SPs must be assigned to stories because stories are what get put into construction through release backlogs.

**Distributing BPs to stories.** Business value considerations should occur at the epic level, not the story level. At the story level, you should consider only how much each story contributes to realizing its epic's estimated benefit (see Figure 4). Why? Because stories specify functionality at a level of detail and granularity that usually makes it difficult to relate to objectives in the business case.<sup>6</sup>

Also, you should keep your expert estimation as local and simple as possible. So, you should consider only one level of the relationship at a time. For epics, look at their relationship to objectives. For stories, look at their relationship to epics.

There are various ways to estimate the stories' proportions of the epics' contribution to benefit; see the sidebar "Assigning Benefit Points to Stories." In Table 3, that task's result

appears in the "story benefit part of epic" column. The resulting portion of the epic's BPs and monetary benefit appear in the "story BPs" and "story benefit" columns immediately to the right.

**Distributing SPs to stories.** Unlike BPs, SPs are commonly assigned directly at the story and task levels, often by new planning-poker sessions by scrum teams. If subcontractors deliver code to the project, they might also use in-house methods for cost estimation. In any event, cost for stories usually isn't estimated by assessing stories' contribution to epics' cost. This is fine because cost estimations are relevant all the way from strategy to construction, especially when subcontractors are involved in construction.

It's still advantageous to express SPs at the story level in terms of their proportion of the epic's SPs (see Figure 4) because this lets you relate them directly to epics. For our example, the stories' proportions of SPs for the epics appear in the "story cost part of epic" column in Table 3, and the resulting portion of the epic's cost appear in the "story SPs" and "story cost" columns to the right.

**Ordering the backlog.** Although Epics 3, 7, and 2 should deliver the most benefit for cost, their stories might not all be as beneficial. In particular, story 7C has an unfortunate benefit/cost ratio and probably shouldn't be put into construction.

The basic principle for ordering the story backlog is straightforward: order it according to the decreasing benefit/cost ratio. If you put stories into construction in that order, the story next in line will always be the one that should generate the most benefit relative to cost. As you put your backlog into construction, if

TABLE 3

Elaborating Epics 3, 7, and 2 into stories for the planned first release.

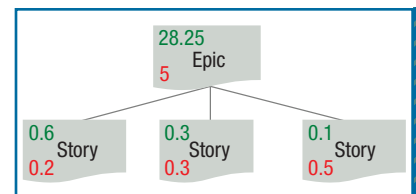
| Epic  | Story | Epic BPs | Epic benefit | Story benefit part of epic | Story BPs | Story benefit | Epic SPs | Epic cost | Story cost part of epic | Story SPs | Story cost | Epic benefit/cost | Story benefit/cost |
|-------|-------|----------|--------------|----------------------------|-----------|---------------|----------|-----------|-------------------------|-----------|------------|-------------------|--------------------|
| 3     | —     | 23.30    | 8.45         | —                          | —         | —             | 3.00     | 1.80      | —                       | —         | —          | 4.69              | —                  |
| —     | 3A    | —        | —            | 0.7                        | 16.31     | 5.91          | —        | —         | 0.6                     | 1.80      | 1.08       | —                 | 5.47               |
| —     | 3B    | —        | —            | 0.3                        | 6.99      | 2.53          | —        | —         | 0.4                     | 1.20      | 0.72       | —                 | 3.52               |
| 7     | —     | 28.25    | 10.24        | —                          | —         | —             | 5.00     | 3.00      | —                       | —         | —          | 3.41              | —                  |
| —     | 7A    | —        | —            | 0.6                        | 16.95     | 6.15          | —        | —         | 0.2                     | 1.00      | 0.60       | —                 | 10.24              |
| —     | 7B    | —        | —            | 0.3                        | 8.48      | 3.07          | —        | —         | 0.3                     | 1.50      | 0.90       | —                 | 3.41               |
| —     | 7C    | —        | —            | 0.1                        | 2.83      | 1.02          | —        | —         | 0.5                     | 2.50      | 1.50       | —                 | 0.68               |
| 2     | —     | 45.13    | 16.36        | —                          | —         | —             | 8.00     | 4.80      | —                       | —         | —          | 3.41              | —                  |
| —     | 2A    | —        | —            | 0.5                        | 22.57     | 8.18          | —        | —         | 0.2                     | 1.60      | 0.96       | —                 | 8.52               |
| —     | 2B    | —        | —            | 0.1                        | 4.51      | 1.64          | —        | —         | 0.2                     | 1.60      | 0.96       | —                 | 1.70               |
| —     | 2C    | —        | —            | 0.2                        | 9.03      | 3.27          | —        | —         | 0.3                     | 2.40      | 1.44       | —                 | 2.27               |
| —     | 2D    | —        | —            | 0.2                        | 9.03      | 3.27          | —        | —         | 0.3                     | 2.40      | 1.44       | —                 | 2.27               |
| 4     | —     | 17.31    | 6.28         | —                          | —         | —             | 5.00     | 3.00      | —                       | —         | —          | 2.09              | —                  |
| 8     | —     | 26.66    | 9.67         | —                          | —         | —             | 8.00     | 4.80      | —                       | —         | —          | 2.01              | —                  |
| 1     | —     | 24.80    | 8.99         | —                          | —         | —             | 8.00     | 4.80      | —                       | —         | —          | 1.87              | —                  |
| 5     | —     | 30.09    | 10.91        | —                          | —         | —             | 13.00    | 7.80      | —                       | —         | —          | 1.40              | —                  |
| 6     | —     | 15.45    | 5.60         | —                          | —         | —             | 13.00    | 7.80      | —                       | —         | —          | 0.72              | —                  |
| Total | —     | 211.00   | 76.50        | —                          | 96.68     | 35.05         | 63.00    | 37.80     | —                       | 16.00     | 9.60       | 2.02              | 3.65               |

you plot the accumulated estimated benefit against the accumulated estimated cost, you get a realization curve with a steep incline easing off, showing how you plan to generate benefit potential faster than you incur cost. Figure 5 shows our example. This maximizes the benefit/cost ratio; that is, it promotes the benefit/cost factor as the most important part of the benefit/cost triangle.

Periodically, you should assess the

information you have available and consider revising your plan. For example, already now, we could plan to drop Story 7C from the current release. If we do so, we'll have capacity in the release to do something more useful.

Suppose we elaborate the next epic (Epic 4) into stories (see Table 4). Instead of spending 1.50 for Story 7C, we can plan to spend 0.90 on Stories 4D and 4A, the two



**FIGURE 4.** The proportions of an epic's BPs and SPs distributed to stories. Green indicates BPs; red indicates SPs. At the story level, you should consider only how much each story contributes to realizing its epic's estimated benefit.

## ASSIGNING BENEFIT POINTS TO STORIES

Benefit estimation for a story should relate to objectives not directly but indirectly, through the benefit points (BPs) of that story's epic. (Epics are high-level requirements perceived at the project's early stages, typically at project inception. In contrast, stories are more refined and elaborated user stories reflecting the project's understanding at more advanced stages.) Previously, we suggested a syntax for epics that explicitly mentioned objectives.<sup>1</sup> To help you to think of stories in terms of their contribution to their epic, you can use the following syntax, which avoids mentioning objectives:

*Story: As <stakeholder A> I can <perform actions d in domain D> by using <functionality f in System S> to <perform actions s in S> in order to <contribute to Epic E>*

Epics can be very high level and represent large chunks of functionality. When distributing BPs from the epic, you might be faced with more stories than you can comfortably keep track of. The solution? Adapt distribution techniques based on assessing the relative importance.

Our analysis of four techniques<sup>2,3</sup> led us to recommend pairwise comparison facilitated by the analytical hierarchy process (AHP),<sup>4</sup> which is easily implemented in a tool. The basis for AHP is that comparing many items in one go is extremely taxing cognitively. Instead, consider only two items (stories) at a time by assessing their relative importance to their epic.

But doesn't this extremely local procedure completely ignore the whole picture and all the relationships between items? Yes and no. From your judgments on two items at a time, AHP deduces a ranking of all the items, even in the face of your inconsistencies. For example, unless you have extraordinary capabilities, your collection of local pairwise comparisons will likely imply that, for some stories *EX* and *EY*, both *EX* > *EY* and *EY* > *EX*. AHP computes a measure of

this inconsistency—the *consistency index* (CI). In line with satisficing rather than optimizing,<sup>5</sup> you can make an educated choice as to a “good enough” CI.

In standard AHP, all possible pairs of stories in an epic must be compared, which can be fine for a moderate number of stories. In our experiments, we implemented a method to reduce the number of required comparisons<sup>6</sup> (with the penalty of having to be more consistent) so that you can use AHP for a large number of stories.

Pairwise comparison is a core element of judgment cognitive processes.<sup>7</sup> Using a method that directly supports your cognitive processes is a good way to obtain better expert estimates.

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stories in Epic 4 with the most efficient benefit/cost ratios. These have a total estimated benefit of 3.14. If only cost was guiding us, we might be tempted to fill the first release's planned capacity with a full 1.50 cost by choosing, say, Stories 4C

and 4D. However, this would just give us the same benefit at a higher cost.

Figure 6 shows the revised plan, which has bumped Story 7C down the line and out of this release and includes Stories 4D and 4A. Table 5

shows the cumulative values of the ordered stories in this revised release. (We discuss the two rightmost columns in Table 5 shortly.)

To utilize the available capacity, we employed a different tactic: maximizing the business value within a

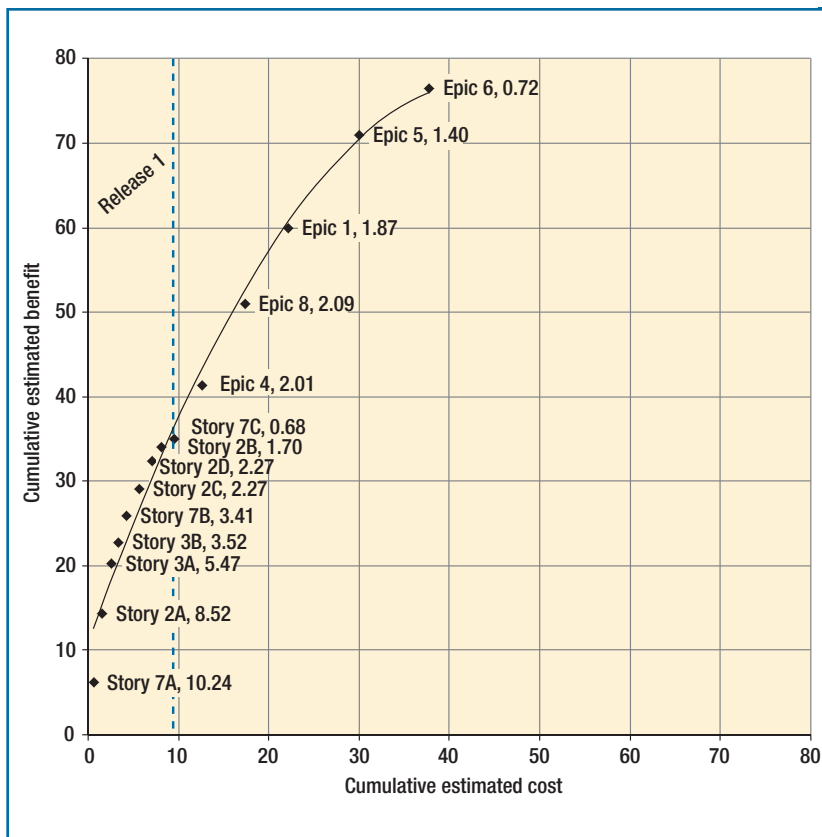
fixed cost. This is a variant of the knapsack problem that's inherent in release planning. So, although you have an overall tactic for maximizing the benefit/cost ratio, you might have to adhere to a fixed cost when adjusting a given release. In this case, you're relating temporarily to the agile triangle rather than the benefit/cost triangle.

Now, to try to optimize your plan at this stage, you could elaborate all the epics, find the stories with the most efficient benefit/cost ratios from the remaining suite of epics, and insert those stories into the available capacity of the release. This would require you to invest more cost earlier—to elaborate the epics—when project experience and knowledge is lower than it might be later. This is, as it sounds, against agile principles. Still, elaboration must occur at some point, and deciding when to do it is a refinement of whatever tactic you're following, which we leave for another discussion.

### Monitoring the Earned Business Value

A common way to measure a project's efficiency is by the metrics of *earned value management* (EVM). EVM requires a means to quantify work done. Agile development accommodates this nicely in its product elements and product backlog. But brace yourself; EVM uses the term "value" in a way that confounds cost with value. This is unfortunate because it supports people's tendency to believe that costlier things have inherently more value.<sup>7</sup>

Anyway, consider the project for some time period  $p$ , at the end of which you've decided to assess project efficiency. The period might represent a sprint, a release, or the entire project up until now. We use these measures:



**FIGURE 5.** The initial plot of the accumulated estimated benefit against the accumulated estimated cost. To the left of the dashed line is the planned realization curve for the release. To the right is the remainder of the queue (unelaborated epics). The numbers next to the epic and story labels are the benefit/cost values.

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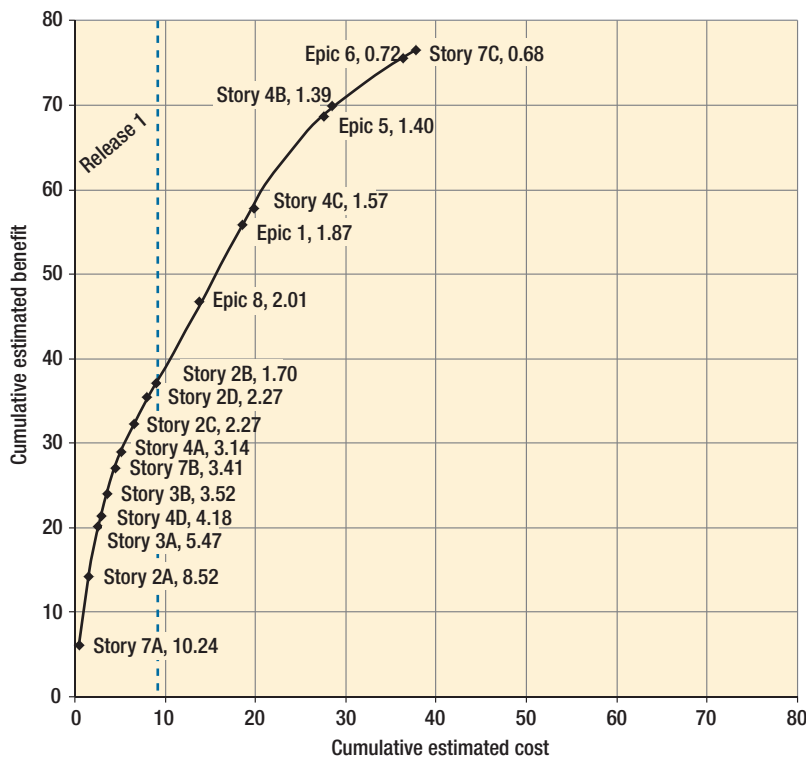
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TABLE 4

Elaborating the next epic, Epic 4, into stories.

| Epic | Story | Epic BPs | Epic benefit | Story benefit part of epic | Story BPs | Story benefit | Epic SPs | Epic cost | Story cost part of epic | Story SPs | Story cost | Epic benefit/cost | Story benefit/cost |
|------|-------|----------|--------------|----------------------------|-----------|---------------|----------|-----------|-------------------------|-----------|------------|-------------------|--------------------|
| 4    | —     | 17.31    | 6.28         | —                          | —         | —             | 5.00     | 3.00      | —                       | —         | —          | 2.09              | —                  |
| —    | 4A    | —        | —            | 0.3                        | 5.19      | 1.88          | —        | —         | 0.2                     | 1.00      | 0.60       | —                 | 3.14               |
| —    | 4B    | —        | —            | 0.2                        | 3.46      | 1.26          | —        | —         | 0.3                     | 1.50      | 0.90       | —                 | 1.39               |
| —    | 4C    | —        | —            | 0.3                        | 5.19      | 1.88          | —        | —         | 0.4                     | 2.00      | 1.20       | —                 | 1.57               |
| —    | 4D    | —        | —            | 0.2                        | 3.46      | 1.26          | —        | —         | 0.1                     | 0.50      | 0.30       | —                 | 4.18               |



**FIGURE 6.** The revised plot of the accumulated estimated benefit against the accumulated estimated cost. To the left of the dashed line is the planned realization curve for the release. To the right is the remainder of the queue. The numbers next to the epic and story labels are the benefit/cost values.

- *Planned value, PV*, is the estimated cost of the user stories planned for completion in  $p$ .
- *Earned value, EV*, is the estimated cost of the user stories actually completed in  $p$ .
- *Actual or adjusted cost, AC*, is the cost of the user stories completed in  $p$ . If your cost estimates are for development only, AC is the actual cost incurred. If your cost estimates are for the lifecycle and you're assuming that the lifecycle cost is proportional to the development cost by a factor of  $L$ , then AC is the actual development cost multiplied by  $L$ . Because AC in the latter case retains an estimate element, we call this the adjusted cost rather than the actual cost.
- *Cost performance index, CPI* =  $EV/AC$ .

Figure 7 illustrates this for our example. The period is the first release, and we planned to produce 10 stories, 7A to 2B, with a total estimated cost of  $PV = 9.00$  (from Table 5). However, the project managed to complete only the first eight stories, 7A to 2C, on time. The total cost

estimate for those eight is  $EV = 6.60$ . At this point, we have the actual development cost for those eight—say, 5.665. Assuming for this example that the lifecycle cost is proportional by a factor of 2 to the development cost, the adjusted cost for the eight stories is  $AC = 11.33$  (from Table 5).

The eight stories' actual cost is more than the estimate for the planned 10 stories. So, the project is both behind schedule and above the planned cost.  $CPI$  ( $6.60/11.33 = 0.58$ ) is well below 1. A  $CPI$  less than 1 indicates that the cost is greater than anticipated. The obvious recommendation for a low  $CPI$  is to take action to improve  $CPI$  in the next period.

As much sense as that makes,  $CPI$  merely measures how much functionality is being produced, not how valuable that functionality is. EVM is designed to support management by the iron triangle. So, let's define measures for business value:

- *Planned business value, PBV*, is the estimated business value of the user stories planned for  $p$ .
- *Earned business value, EBV*, is the estimated business value of the user stories completed in  $p$ .
- *Adjusted benefit, AB*, is the reestimated benefit for the user stories completed in  $p$ .
- *Benefit performance index, BPI* =  $EBV/AB$ .
- *Benefit cost performance index, BCPI* =  $EBV/AC$ .
- *Adjusted benefit cost, ABC* =  $AB/AC$ .

In our example,  $PBV = 37.17$  and  $EBV = 32.26$  (from Table 5). Although  $EBV$  is less than  $PBV$ ,  $BCPI$  ( $32.26/11.33 = 2.85$ ) is well above 1. A  $BCPI$  less than 1 means you're investing more money than you expect to gain, so you should consider

TABLE 5

The stories' revised first release, sorted by the decreasing benefit/cost ratio.

| Story | Cumulative results |         |       |      |                  |               |
|-------|--------------------|---------|-------|------|------------------|---------------|
|       | BPs                | Benefit | SPs   | Cost | Adjusted benefit | Adjusted cost |
| 7A    | 16.95              | 6.15    | 1.00  | 0.60 | 6.15             | 1.20          |
| 2A    | 39.52              | 14.33   | 2.60  | 1.56 | 10.24            | 2.54          |
| 3A    | 55.83              | 20.24   | 4.40  | 2.64 | 16.15            | 2.87          |
| 4D    | 59.29              | 21.50   | 4.90  | 2.94 | 17.40            | 3.77          |
| 3B    | 66.28              | 24.03   | 6.10  | 3.66 | 19.94            | 5.21          |
| 7B    | 74.75              | 27.10   | 7.60  | 4.56 | 23.01            | 6.11          |
| 4A    | 79.95              | 28.99   | 8.60  | 5.16 | 24.89            | 7.01          |
| 2C    | 88.97              | 32.26   | 11.00 | 6.60 | 28.17            | 11.33         |
| 2D    | 98.00              | 35.53   | 13.40 | 8.04 | —                | —             |
| 2B    | 102.51             | 37.17   | 15.00 | 9.00 | —                | —             |

alternative investments. A cleverly prioritized project will start with a high  $BCPI$ , earning much business value compared to the cost expended at the project's start.

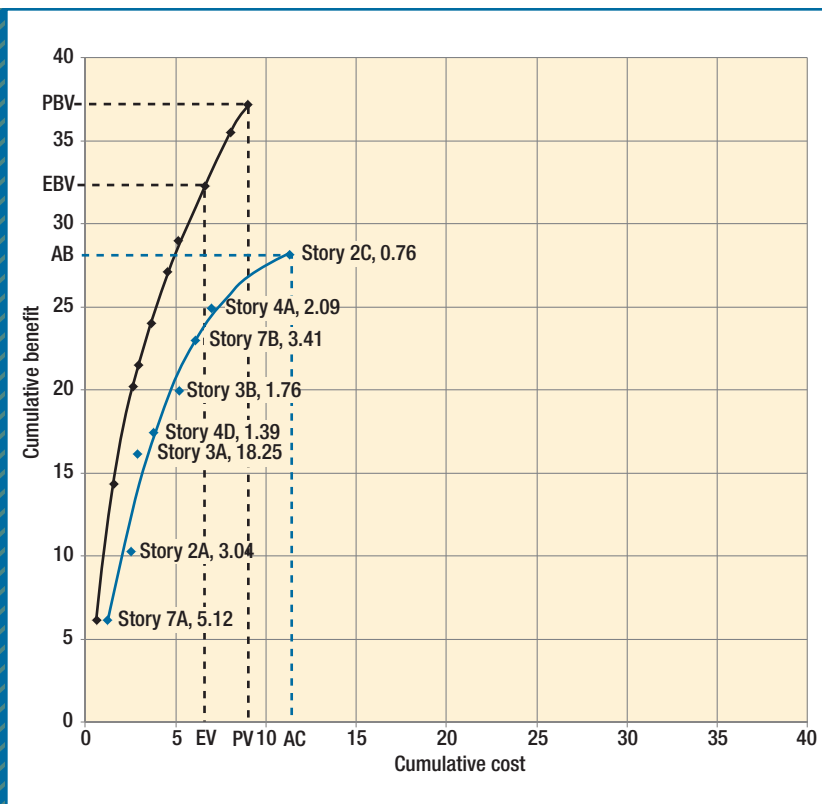
After the first release in our example, the project's velocity regarding cost isn't good, but its velocity regarding business value is acceptable relative to the cost. Such balanced information is important when you're reporting to project management and project or product owners. It's also important for virtually every project stakeholder because it gives a wider picture that includes the progress in terms of value for the customer, in addition to the amount of functionality.

In traditional EVM,  $AC$  is the development expenditure. As we mentioned before, we've generalized  $AC$  to account for the postdeployment cost, which hasn't yet been incurred. For our *earned business value*

*management* (EBVM) regime, we define the analogous  $AB$ .  $AB$  is based on experience from using increments deployed from the project or from other reestimates of benefit due to, for example, changes in external factors such as legislation or dependencies on other systems' evolution.

For our example, imagine that Story 2A was found to be overrated once the stakeholders saw its functionality in action and was subsequently reestimated to half its original benefit. This gives  $AB = 28.17$  (from Table 5), which gives  $BPI = 32.26/28.17 = 1.15$ .  $BPI$  is a pure business value metric, and values greater than 1 mean you're generating less business value than you expected. Still,  $ABC = 28.17/11.33 = 2.49$ , so we're actually faring quite well.

You can derive several other metrics from the basic ones of EVM and EBVM. You can now construct your own dashboard for monitoring



**FIGURE 7.** The planned realization curve (black) and actual realization curve (blue). The numbers next to the story labels are the AB/AC (adjusted benefit / adjusted cost) values for each story. PBV is the planned business value, EBV is the earned business value, EV is the earned value, and PV is the planned value.

project efficiency in terms of cost and benefit. As you can see, BPs and SPs are at the core of how we define the EVM and EVBM metrics here. In fact, it can be advantageous to make BPs and SPs even more explicit; see the sidebar “Agile EBVM in Practice.” BPs and SPs give you means to define a host of metrics that tap directly into your construction line while giving meaningful indications in terms of the business case.

## Dependencies

Functional, temporal, and architectural dependencies between product elements are common. In addition, worldly factors such as available

expertise, illness, conflicts, and external constraints might influence when stories are put into construction.

We don’t discuss such dependencies here. Also, it’s important to realize that a backlog that’s perfectly ordered regarding the benefit/cost ratio is an input to the release-planning stage, which fully deals with dependencies. Our approach is integral to more detailed dependency handling. For example, Mark Denne and Jane Cleland-Huang gave a thorough account of dependencies’ consequences on cost and benefit realization.<sup>8</sup> They presented a heuristic that approximates the optimal ordering of dependency-heavy product elements

with respect to return on investment in the net-present-value regime. Assigning points to product elements would provide the necessary cost and benefit estimates before such heuristics are applied. Owing to dependencies, your backlog might not be perfectly ordered regarding the benefit/cost ratio. However, because you assign BPs and SPs, you can keep track of the project’s planned and actual productivity even in the turmoil of dependency-driven release planning.

That said, we claim that dependencies can also result from unhealthy architectural work and from dividing functionality into pieces that don’t make operational sense. The focus on organizational agility has brought forth concepts such as the minimum marketable feature, minimum business increment, and minimum viable product. These concepts embody minimal product elements that provide added value for the customer. The flip side is that a product element that’s involved in dependencies doesn’t bring value in and of itself.

Furthermore, the present-day focus on capabilities and services<sup>9,10</sup> stresses the development of independent pieces of functionality that persist over time, in multiple contexts at both the business and technical levels.<sup>11</sup> If you’re in line with these architectural modes, when strong dependencies arise, you can reconsider how to divide your functionality. For example, product elements with strong dependencies might more sensibly be combined into one element.

**W**ith both BPs and SPs in your vocabulary, you can enhance your



## AGILE EBVM IN PRACTICE

When applying *earned business value management* (EBVM), we've found it useful to relate to alternative, but equivalent, expressions for the *cost performance index* (*CPI*) and *benefit cost performance index* (*BCPI*) that clearly separate points and monetary value. (For more on *CPI* and *BCPI*, see the main article.) Although these expressions are less streamlined in definition than those in the main article, our experience is that project stakeholders intuitively understand these metrics better and that these metrics increase the project state's transparency. They stimulate you to use EBVM purely on the basis of points and monetary values that might be more accessible, such as total budgeted cost and benefit. We've found the effort required to collect data and calculate the metrics to be almost negligible.

You'll likely need to try this out in practice—for example, in a spreadsheet to come to grips with the larger number of expressions. But once you've done that, you might find this way simpler.

For a period  $p$ , consider

- $PSP$ , the planned story points (SPs);
- $ESP$ , the earned SPs;
- $TSP$ , the total number of SPs assigned in the project; and
- $FSP = ESP/TSP$ , the proportion of  $TSP$  that's earned.

In our example, the period in question is the first (revised) release (see Table 5 and Figure 7 in the main article);  $PSP = 15$ ,  $ESP = 11$ , and  $TSP = 63$ . Then,  $FSP = 11/63 = 17.50$  percent.

Furthermore, consider

- $VSP$ , the monetary value of an SP you're using for  $p$ ;
- $TPV = TSP \times VSP$ , the estimated total lifecycle cost—or total planned value—given  $VSP$ ; and
- $FC = AC/TPV$ , the proportion of  $TPV$  that's committed, where  $AC$  is the adjusted cost (which takes into account the actual development cost and the estimated lifecycle cost of the completed user stories; see the main article).

In our example,  $VSP = 0.60$  million Norwegian kroner (US\$1 = 8.5 kr),  $TPV = 37.80$  million kr, and  $FC = 11.33/37.80 = 29.97$  percent. With simple math, you'll verify that  $CPI = FSP/FC$ .

For our example,  $CPI = 17.50/29.97 = 0.58$ —the same as we calculated the standard way in the main article.

Now, consider

- $PBP$ , the planned benefit points (BPs);
- $EBP$ , the earned BPs;
- $TBP$ , the total number of BPs assigned in the project; and
- $FBP = EBP/TBP$ , the proportion of  $TBP$  that's earned.

In our example (see Table 5 and Figure 7 in the main article),  $PBP = 102.51$ ,  $EBP = 88.97$ , and  $TBP = 211$ . Then,  $FBP = 88.97/211 = 42.17$  percent.

Furthermore, consider

- $VBP$ , the monetary value of a BP you're using for  $p$ ;
- $TPBV = TBP \times VBP$ , the estimated total lifecycle benefit—or total planned business value—given  $VBP$ ; and
- $FB = AB/TPBV$ , the proportion of  $TPBV$  that's committed, where  $AB$  is the adjusted benefit (the reestimated benefit for the completed user stories).

In our example,  $VBP = 0.36$  million,  $TPBV = 76.50$  million, and  $FB = 28.17/76.50 = 36.82$  percent. We can easily verify that the benefit performance index  $BPI = FBP/FB$  and that  $BCPI = FBP/FC \times TPBV/TPV$ . (For more on  $BPI$ , see the main article.) For our example,  $BPI = 42.17/36.82 = 1.15$  and  $BCPI = 42.17/29.97 \times 76.50/37.80 = 1.41 \times 2.02 = 2.85$ —the same as we calculated the standard way in the main article.

Some of the cost metrics ( $PSP$ ,  $ESP$ ,  $TSP$ ,  $VSP$ , and  $TPV$ ) are in line with other frameworks.<sup>1–3</sup>  $VSP$  is often understood as the budgeted baseline cost per SP fixed throughout the project. But you can choose to have a dynamic  $VSP$  reflecting project experience, depending on what kind of “agile” you're committed to. You can even equip your dashboard with a fixed  $VSP$  and a dynamic  $VSP'$ . The same goes for  $VBP$ , of course.

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capability to systematize project knowledge and project learning on the aspects that matter the most: those of business value. Here, we showed how to order a product backlog and keep track of productivity in terms of both cost and business value. In that light, it's pertinent to ask how anyone could think of running projects aimed at

providing value for the customer on the basis of cost metrics alone. In another discussion, we'll show how our approach can incorporate risk and uncertainty. 🌀

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